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Correlation and predictive value of novel anthropometric indicators with adult sarcopenia and sarcopenia obesity

Xue Wu[1,2], Ziang Chen[1,2], Yuanhui Zhao[1] & Hong Ren[1]✉

The present study leverages the comprehensive data from the National Health and Nutrition Examination Survey (NHANES) to examine the Influencing factors of sarcopenia (SA) and sarcopenic obesity (SO). The investigation is designed to a non-invasive, cost-effective, and convenient method that is applicable to the adult population, enabling the accurate and simultaneous detection of risks associated with SA and SO. Furthermore, this research will evaluate the critical values of effective anthropometric indicators, providing early warning for risk management in self-health care and offering valuable insights for subsequent research and clinical practice. The data pertaining to NHANES participants were meticulously selected from the databases of six cycles, spanning from 2001 to 2004 and 2011 to 2018. Utilizing the diagnostic criteria established by the American Foundation for the National Institutes of Health (FNIH), anthropometric measurement data were extracted to construct composite indices. These indices were then cross-referenced with diagnostic assessments from dual-energy X-ray absorptiometry and bioelectrical impedance analysis to examine the correlations between various metrics and the incidence of diseases. R software (version 4.3.3) was used for analysis, and the primary analytical methods employed included logistic regression, restricted cubic splines (RCS), and the Receiver Operating Characteristic (ROC) curve analysis (AUC). Sarcopenia and sarcopenic obesity are commonly observed in individuals within the middle-aged and elderly demographics. The prevalence of these conditions is higher among middle-aged men of a given age when contrasted with women at the age of 40. All anthropometric indexes demonstrated a positive correlation with the onset of SA and SO, with the association with waist-to-height ratio (WHtR) showcasing a heightened strength subsequent to the adjustment for all covariates. The predictive models of all ROC curves performed commendably, particularly with the body roundness index and WHtR forecasting models exhibiting superior performance, the area under the AUC curve is 0.87 (95 CI% 0.85, 0.88) and 0.86 (95 CI% 0.85, 0.88), respectively. The RCS curve delineated a distinctive J-shaped distribution for each physical index in concurrence with SA and SO, signifying an optimal value at which the incidence of these conditions is minimal; conversely, deviations from this optimal value entailed an escalated risk of disease. Diverse anthropometric index metrics bear a strong correlation with adult onset of sarcopenia and sarcopenic obesity, each displaying commendable predictive capability. Notably, the body roundness index and waist-to-height ratio may harbor heightened potential as indicative anthropometric indexes. Furthermore, the dose-effect relationship analysis inferred that the lowest disease risk is manifested among individuals with specific index profiles, thereby advocating for autonomous health monitoring to promote physical activity and bolster nutrient intake, thus mitigating the risk of sarcopenia and sarcopenic obesity.

Keywords Anthropometric indicator, Sarcopenia, Sarcopenic obesity

Sarcopenia (SA), a multifaceted syndrome, increases the risk of falls, fractures, and impaired daily activities[1,2]. It is further correlated with elevated overall mortality rates, an increased susceptibility to depression, and a spectrum of metabolic disorders[3], exerting a profound impact on the individual, community, and economy[4]. The concurrent occurrence of excessive adiposity and diminished muscle mass is designated as sarcopenic obesity

[1]Beijing Sport University, Beijing, China. [2]Wu Xue and Chen Ziang contributed equally to this work. ✉email: renhong@bsu.edu.cn

(SO)[5], a complication emanating from the amalgamation of sarcopenia and obesity[6]. Sarcopenic obesity is closely associated with frailty, compromised cardiovascular metabolic function, physical disability, and mortality[7].

Presently, conventional modalities for diagnosing skeletal muscle wasting encompass magnetic resonance imaging (MRI) and computed tomography (CT), methodologies that provide precise direct observations, necessitate the utilization of costly equipment and incur significant evaluation costs. Bioelectrical impedance analysis (BIA) stands out as a prevalent evaluation technique[8,9], capable of delivering relatively accurate results with relative ease. However, it remains susceptible to a variety of influences, including electrode placement, variations in testing positions, and changes in body fluid composition[10]. Additionally, discrepancies in outcomes may arise due to variations among different device models[11]. Dual-energy X-ray absorptiometry (DEXA) is extensively acknowledged as the prevalent method for the assessment of SA and SO, lauded for its cost-effectiveness and considerable precision[12]. Nevertheless, the necessity for a certain level of equipment proficiency poses a limitation on its application, thereby restricting the capacity for self-monitoring and the evaluation of daily living activities within populations at elevated risk. Beyond the application of equipment, the diagnostic process for sarcopenia frequently involves the use of various questionnaires, including but not limited to SARC-F, SARC-CalF, MSRA-5, and MSRA-7. However, the existing literature displays a notable lack of consensus regarding the validity of these questionnaires. Furthermore, the process of questionnaire completion may be prone to biases introduced by subjective factors[13,14].

In light of the widening chasm in health disparities, the restricted access to medical care, and the unrelenting escalation of healthcare costs[15], the practice of self-health management has emerged as a promising intervention strategy[16]. This entails individuals acquiring the competencies necessary for monitoring their own health status and utilizing behavioral goals as motivational drivers[17]. A robust correlation exists between anthropometric indices and behaviors pertinent to self-health management[18], the outcomes of measurements contribute significantly to the facilitation of self-management practices[19]. Anthropometric indices are synthesized into specific metrics through formulaic calculations, providing a delineation of an individual's relative physical health status. The principal advantage of this method lies in its measurement simplicity[20], in addition to the fact that numerous studies affirming the value of anthropometric indices in forecasting conditions such as obesity, cardiovascular disease, and nephrolithiasis[21–24], as well as in discerning the association between these indices and overall mortality rates[25].

The calf circumference (CC) metric is widely utilized as a convenient and relatively precise indicator in the assessment of elderly patients' sarcopenia[26]. However, due to the potential for obesity to distort limb measurements, the accurate application of the CC index necessitates adjustment in relation to Body Mass Index (BMI)[27]. This adjustment may diminish the predictive value in patients with sarcopenic obesity, potentially leading to misdiagnoses that could exacerbate the progression of sarcopenic obesity and precipitate the onset of comprehensive sarcopenia[28]. Commonly used anthropometric indicators, such as body weight and BMI, are relatively straightforward to calculate but are limited in their ability to detect overweight populations[29–31]. Utilizing these as the sole criteria for assessing obesity may lead to misclassification of individuals with occult obesity or those with resilient physiologies[32], thereby underestimating the health risks associated with visceral adiposity[33,34]. A considerable proportion of the elderly exhibit a normal BMI, yet may be afflicted by varying degrees of sarcopenic obesity prior to the manifestation of pronounced muscle atrophy[35]. Consequently, it is essential to appraise a range of anthropometric indicators and to elucidate their correlations with disease risk[36]. Research indicates that indices such as the Body Roundness Index (BRI), the z-score of the log-transformed A Body Shape Index (LBSIZ), the A Body Shape Index (ABSI), and the Waist-to-Height Ratio (WHtR) are instrumental in the detection of SA or SO[37–40]. Despite evidence linking anthropometric indicators to specific metabolic ailments[41], the relationship between these indicators and sarcopenia, as well as sarcopenic obesity, necessitates further investigation. Moreover, a consensus on the diagnostic criteria for the condition remains elusive[42]. A comprehensive comparison across a substantial dataset should be conducted to identify the optimal solution.

The present study leverages the comprehensive data from the National Health and Nutrition Examination Survey (NHANES) to examine the Influencing factors of SA and SO. The investigation is designed to a non-invasive, cost-effective, and convenient method that is applicable to the adult population, enabling the accurate and simultaneous detection of risks associated with SA and SO. Furthermore, this research will evaluate the critical values of effective anthropometric indicators, providing early warning for risk management in self-health care and offering valuable insights for subsequent research and clinical practice.

Methods

Study participants

This study is a cross-sectional study, and the data is derived from the US NHANES database (https://www.cdc.gov/nchs/nhanes/index.htm). The data in this database comes from a nationwide health survey conducted annually, using a complex, multi-stage probability sampling method that can represent the health and nutritional status of the population[43]. All participants signed written informed consent forms.

Age-related aging is often considered a cause of sarcopenia (SA) and osteopenia (SO). Studies have shown that factors such as exercise, chronic inflammation, and nutritional status also affect the onset of SA and SO[44,45]. Due to the limitations of the measurement content of the database in different years, this study extracted DEXA measurement data and relevant covariates from the NHANES database from 2011 to 2018 as the main research objects, and used relevant data measured by BIA in the 2001–2004 database for index verification. After screening the data from 2011 to 2018, excluding individuals under 20 years of age and those with missing data, this study finally included 8159 participants (Fig. 1).

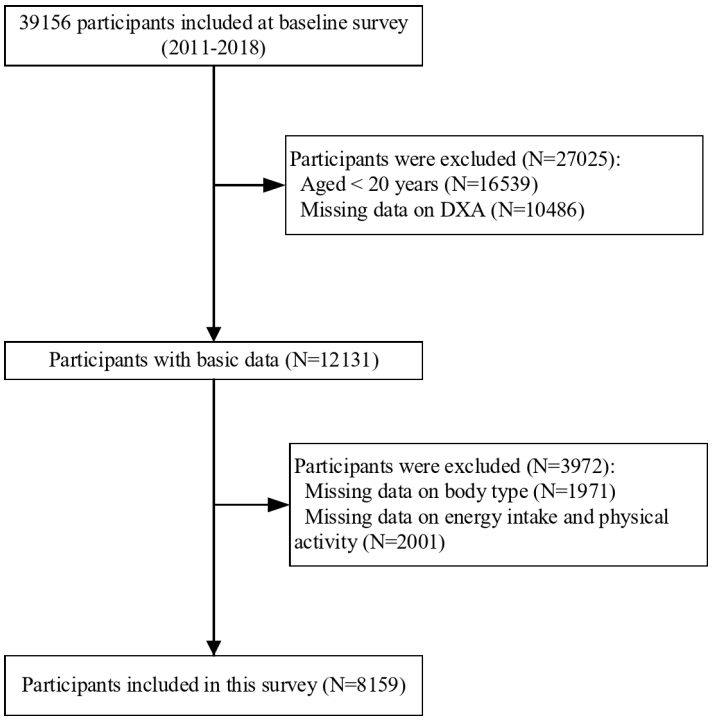


Fig. 1. Flow chart of the study population.

Measurements

Disease threshold measurement

The data utilized in this study are derived from the National Health and Nutrition Examination Survey in the United States. Consequently, the racial characteristics are assessed in accordance with the standards established by the American Foundation for the National Institutes of Health (FNIH)[46]. The diagnosis of sarcopenia is established when the total skeletal muscle mass/body mass index (BMI) measured by dual-energy X-ray falls below 0.789 kg/BMI in men and below 0.512 kg/BMI in women. If the criteria for sarcopenia are met and BMI is equal to or exceeds 30, it is diagnosed as sarcopenic obesity[5].

Furthermore, bioelectrical impedance analysis (BIA) can be utilized to estimate sarcopenia[47]. In line with the equipment requisites of the 2001–2004 NHANES survey, regression equations are employed to calculate skeletal muscle mass (SM) using resistance values[48]:

$$SM = \left[\frac{height\,(cm)^2}{resistance\,(\Omega)\times\ 0.401} + (gender\times\ 3.825) + (age\times\ (-0.071))\right] + 5.102.$$

Subsequently, the estimated SM is inputted into the equation to compute the skeletal muscle mass index (SMI), with the high-risk threshold for sarcopenia identified as 8.50 kg/m^2 for men and 5.75 kg/m^2 for women[49].

Anthropometric index measurement

The requisite data for computing diverse anthropometric parameters encompass measurements such as waist circumference, height, and weight, delineated in a multitude of units. The computational algorithms and units pertinent to assorted physical indices are delineated in Table 1.

Other indicators' measurement

The nutrient intake data were collected via in-person interviews. All dietary interviewers underwent a rigorous week-long training program and completed practical exercises prior to conducting independent interviews. The specific procedures followed the "NHANES Dietary Recall Interview Measurement Guidelines". Study participants were interviewed on consecutive days, and the data used in the study represent their daily average energy and protein intake.

Information on physical inactivity and sedentary behavior was obtained through interviews and questionnaires based on the World Health Organization's Global Physical Activity Questionnaire (GPAQ). Interviewers utilized computer-assisted personal interview systems to survey participants at their homes, inquiring about their typical daily sedentary time. There is a linear correlation between sitting time and the risk of metabolic syndrome[51], according to the SIT-ACT risk matrix[52], sedentary behavior data were extracted and standardized into time units (hours), and then categorized into four levels: less than 4 h, 4–6 h, 6–8 h, and more than 8 h.

Anthropometric index	Calculation formula
Waist-height ratio (WHtR)	$WHtR = \frac{waist(\mathrm{m})}{height(\mathrm{m})}$
Body mass index (BMI)	$BMI = \frac{weight(\mathrm{kg})}{height(\mathrm{cm})^2}$
A body shape index (ABSI)	$ABSI = \frac{waist(\mathrm{m})}{BMI(\mathrm{kg/m^2})^{\frac{2}{3}} \times height(\mathrm{m})^{\frac{1}{2}}}$
Body roundness index (BRI)	$BRI = 364.2 - 365.5 \times \sqrt{1 - \left(\frac{(waist\ (\mathrm{m})/2\pi)^2}{(0.5 \times height\ (\mathrm{m}))^2}\right)}$
The z-score of the log-transformed ABSI[50] (LBSIZ)	$LBSI = \log(ABSI) \quad LBSIZ = \frac{LBSI - LBSI(mean)}{LBSI(s.d.)}$

Table 1. Novel anthropometric index calculation formula.

Demographic variables, including age, sex, race, education level, and poverty level, were collected by trained interviewers at participants' residences using computer-assisted personal interview systems. Relevant demographic data were extracted for the study, and a sophisticated weighted process was applied to the overall data using sample weight.

Statistical analysis

In the investigation, R software (version 4.3.3) was employed for statistical analysis, wherein all analyses were appropriately weighted using the survey package. Categorical variables were presented as frequencies in percentages, while continuous variables were represented as means along with their 95% confidence intervals. To examine the association between different body mass index categories and adult sarcopenia, as well as sarcopenic obesity, several logistic regression models were established, including Model 1, Model 2, and Model 3, which were adjusted for various covariates.

Furthermore, the rms package within the R software facilitated the execution of restricted cubic spline (RCS) analysis, thereby elucidating the dose-response relationship between body measurement indicators and the risk of adult sarcopenia and sarcopenic obesity. The analysis involved comparison of subgroups by sex, with the number of nodes being set to 4 based on percentiles (P_5, P_{25}, P_{75}, P_{95})[53]. Post stepwise regression selection, models predicting sarcopenia and sarcopenic obesity were formulated utilizing diverse body measurement indicators. Subsequently, the pROC and ggplot2 packages were utilized to craft receiver operating characteristic (ROC) curves.

Moreover, the data underwent division into a training set and a validation set at an 8:2 ratio to verify the stability of the model. Lastly, the data measured by bioelectrical impedance in the NHANES database from 2001 to 2004 was re-fitted to authenticate the predictive findings.

Results

Baseline data distribution

Population Representation and Demographics This investigation encompassed a cohort of 8159 individuals and, through utilizing a meticulously calculated complex weighted population estimate, is anticipated to mirror the characteristics of 213,963,585 U.S. residents aged 20 years and above. The mean age of the aforementioned 8159 participants was recorded as (39.40 ± 11.53) years. The sex distribution of the participants is equitable, with individuals of non-Hispanic white descent constituting the preponderance (63%) of the study population. Furthermore, a majority of participants boast a high school education or above (67%), while the prevalence of sedentary behavior exceeding 8 h is notably pervasive (41%). Examination focusing on groups afflicted by sarcopenia and sarcopenic obesity unveiled a predilection for these maladies among older age cohorts, with a slightly elevated prevalence of these conditions witnessed among male participants in comparison to their female counterparts, as explicated in Table 2.

Associations and Correlations Employing normative adults and individuals afflicted by sarcopenia as comparative reference points, logistic regression analysis accounting for adjusted weights revealed a significant correlation between advanced age, race, socioeconomic status, educational attainment, hyperglycemia, physical inactivity, caloric intake, and protein consumption with the occurrence of sarcopenia and sarcopenic obesity, all attaining statistical significance ($P < 0.05$). Conversely, no statistically significant correlations were discerned in relation to sex, hypertension, hypercholesterolemia, and prolonged sedentary behavior ($P > 0.05$) (Table 2).

Correlation analysis of various anthropometric indices

Controlled for various covariates, three regression models were formulated, and collinearity diagnostics revealed the absence of multicollinearity issues (VIF < 5). The outcomes of the univariate logistic regression model (Model 1) exhibited a positive correlation between all metrics and the overall incidence rates of SA and SO, and were of statistical significance ($P < 0.05$). Notably, the association between body mass index and waist-to-height ratio displayed heightened strength. Upon adjusting for demographic variables (Model 2), all metrics sustained their positive correlation with the incidence rates of SA and SO, retaining statistical significance ($P < 0.05$). The association strength of the waist-to-height ratio was notably robust. Comparable findings were derived after controlling for all covariates (Table 3).

Variable	Sarcopenia				Sarcopenic obesity			
	Overall $N=8159$ (100%)[1]	Yes $N=695$ (7.3%)[1]	No $N=7464$ (93%)[1]	P[2]	Overall $N=8159$ (100%)[1]	Yes $N=484$ (5.5%)[1]	No $N=7675$ (95%)[1]	P[2]
Sex				0.074				0.3
Male	3932 (49%)	339 (54%)	3593 (49%)		3932 (49%)	220 (52%)	3712 (49%)	
Female	4227 (51%)	356 (46%)	3871 (51%)		4227 (51%)	264 (48%)	3963 (51%)	
Age (years)				< 0.001				< 0.001
20–34	3102 (37%)	179 (28%)	2923 (38%)		3102 (37%)	126 (28%)	2976 (38%)	
35–44	2009 (24%)	142 (21%)	1867 (24%)		2009 (24%)	104 (22%)	1905 (24%)	
45–54	2048 (25%)	223 (27%)	1825 (25%)		2048 (25%)	148 (25%)	1900 (25%)	
> 55	1000 (14%)	151 (24%)	849 (13%)		1000 (14%)	106 (25%)	894 (13%)	
Race				< 0.001				< 0.001
Mexican American	1136 (9.5%)	232 (24%)	904 (8.4%)		1136 (9.5%)	169 (24%)	967 (8.7%)	
Other Hispanic	796 (6.8%)	107 (12%)	689 (6.4%)		796 (6.8%)	75 (11%)	721 (6.5%)	
Non-Hispanic White	3006 (63%)	202 (51%)	2804 (64%)		3006 (63%)	154 (54%)	2852 (64%)	
Non-Hispanic Black	1747 (11%)	47 (3.8%)	1700 (12%)		1,747 (11%)	45 (4.8%)	1702 (12%)	
Other race	1474 (9.1%)	107 (9.3%)	1367 (9.1%)		1,474 (9.1%)	41 (6.5%)	1433 (9.3%)	
Poverty level	3.00 (1.44, 5.00)	2.00 (1.08, 3.79)	3.09 (1.48, 5.00)	< 0.001	3.00 (1.44, 5.00)	2.06 (1.16, 3.82)	3.07 (1.47, 5.00)	< 0.001
Education level				< 0.001				< 0.001
Less than high school education	1301 (11%)	195 (22%)	1106 (11%)		1301 (11%)	137 (20%)	1164 (11%)	
High school graduate	1749 (21%)	185 (28%)	1564 (21%)		1749 (21%)	130 (28%)	1619 (21%)	
High school education or above	5109 (67%)	315 (50%)	4794 (69%)		5109 (67%)	217 (52%)	4892 (68%)	
Hyperglycemia				0.005				0.007
No	7868 (97%)	653 (94%)	7215 (97%)		7868 (97%)	451 (94%)	7417 (97%)	
Yes	291 (3.3%)	42 (5.8%)	249 (3.1%)		291 (3.3%)	33 (6.2%)	258 (3.2%)	
Hypertension				0.6				0.5
No	8068 (99%)	682 (99%)	7386 (99%)		8068 (99%)	476 (99%)	7592 (99%)	
Yes	91 (1.3%)	13 (1.1%)	78 (1.3%)		91 (1.3%)	8 (0.9%)	83 (1.3%)	
High cholesterol				0.10				0.5
No	7967 (98%)	662 (97%)	7305 (98%)		7967 (98%)	463 (97%)	7504 (98%)	
Yes	192 (2.3%)	33 (3.2%)	159 (2.2%)		192 (2.3%)	21 (2.7%)	171 (2.3%)	
Physical inactivity				0.015				0.004
No	7809 (95%)	643 (93%)	7166 (96%)		7809 (95%)	441 (92%)	7368 (96%)	
Yes	350 (4.6%)	52 (7.0%)	298 (4.4%)		350 (4.6%)	43 (8.1%)	307 (4.3%)	
Sedentary time (hours)				0.5				0.6
< 4	1928 (21%)	186 (23%)	1742 (20%)		1928 (21%)	129 (23%)	1799 (20%)	
4–6	1883 (23%)	176 (23%)	1707 (23%)		1883 (23%)	114 (21%)	1769 (23%)	
6–8	1214 (15%)	103 (15%)	1111 (15%)		1214 (15%)	73 (15%)	1141 (15%)	
> 8	3134 (41%)	230 (38%)	2904 (41%)		3134 (41%)	168 (40%)	2966 (41%)	
Calorie intake (kcal/day)	2048 (1592, 2596)	1873 (1474, 2467)	2063 (1599, 2605)	< 0.001	2048 (1592, 2596)	1876 (1490, 2488)	2060 (1597, 2601)	0.013
Protein intake (g/day)	79 (60, 103)	73 (55, 97)	80 (60, 103)	0.002	79 (60, 103)	73 (55, 98)	80 (60, 103)	0.014
Waist-height ratio (WHtR)	0.57 (0.51, 0.64)	0.69 (0.63, 0.75)	0.56 (0.50, 0.62)	< 0.001	0.57 (0.51, 0.64)	0.72 (0.68, 0.78)	0.56 (0.50, 0.62)	< 0.001
Body roundness index (BRI)	4.70 (3.53, 6.28)	7.66 (6.12, 9.29)	4.55 (3.43, 5.97)	< 0.001	4.70 (3.53, 6.28)	8.34 (7.29, 10.02)	4.58 (3.45, 5.97)	< 0.001
Body mass index (BMI)	27.7 (24.0, 32.4)	34.2 (30.1, 39.4)	27.3 (23.7, 31.7)	< 0.001	27.7 (24.0, 32.4)	36.5 (33.1, 41.5)	27.3 (23.7, 31.6)	< 0.001
A body shape index (ABSI)	0.081 (0.078, 0.083)	0.082 (0.080, 0.085)	0.080 (0.077, 0.083)	< 0.001	0.081 (0.078, 0.083)	0.082 (0.079, 0.085)	0.080 (0.077, 0.083)	< 0.001
Z-score of the log-transformed ABSI (LBSIZ)	− 595 (− 604, − 587)	− 590 (− 598, − 582)	− 595 (− 604, − 588)	< 0.001	− 595 (− 604, − 587)	− 591 (− 599, − 582)	− 595 (− 604, − 587)	< 0.001

Table 2. Baseline characteristic of the included participants (2011–2018 NHANES database). [1]Median (IQR) for continuous; n (%) for categorical. [2]Chi-squared test with Rao and Scott's second-order correction; Wilcoxon rank-sum test for complex survey samples.

Index		Model 1	Model 2	Model 3
Sarcopenia	ABSI	3.17 (2.43, 4.12)*	2.34 (1.74, 3.17)*	2.34 (1.71, 3.19)*
	BRI	1.57 (1.51, 1.63)*	1.66 (1.58, 1.74)*	1.67 (1.59, 1.76)*
	BMI	1.13 (1.12, 1.15)*	1.15 (1.13, 1.17)*	1.15 (1.14, 1.17)*
	WHtR	3.30 (2.99, 3.64)*	3.85 (3.40, 4.36)*	3.93 (3.45, 4.48)*
	LBSIZ	1.04 (1.03, 1.05)*	1.03 (1.02, 1.04)*	1.03 (1.02, 1.04)*
Sarcopenic Obesity	ABSI	2.48 (1.85, 3.32)*	1.79 (1.29, 2.49)*	1.78 (1.27, 2.49)*
	BRI	1.76 (1.68, 1.85)*	1.89 (1.78, 2.00)*	1.91 (1.79, 2.03)*
	BMI	1.18 (1.17, 1.20)*	1.21 (1.19, 1.23)*	1.21 (1.19, 1.24)*
	WHtR	4.67 (4.13, 5.29)*	5.73 (4.91, 6.69)*	5.89 (5.02, 6.92)*
	LBSIZ	1.03 (1.02, 1.04)*	1.02 (1.01, 1.03)*	1.02 (1.01, 1.03)*

Table 3. Logistic regression of anthropometric indicators with SA and SO [OR (95% CI)] Model 1 is a univariate model. Model 2 adjusts for demographic variables, including age, sex, race, poverty level and education level. Model 3 includes all relevant covariates, including demographic variables, hyperglycemia, hypertension, high cholesterol, physical inactivity, sedentary time, and dietary intakes of calories and proteins. *ABSI* a body shape index, *BRI* body roundness index, *BMI* body mass index, *WHtR* waist-height ratio, *LBSIZ* the z-score of the log-transformed ABSI. *$P < 0.05$.

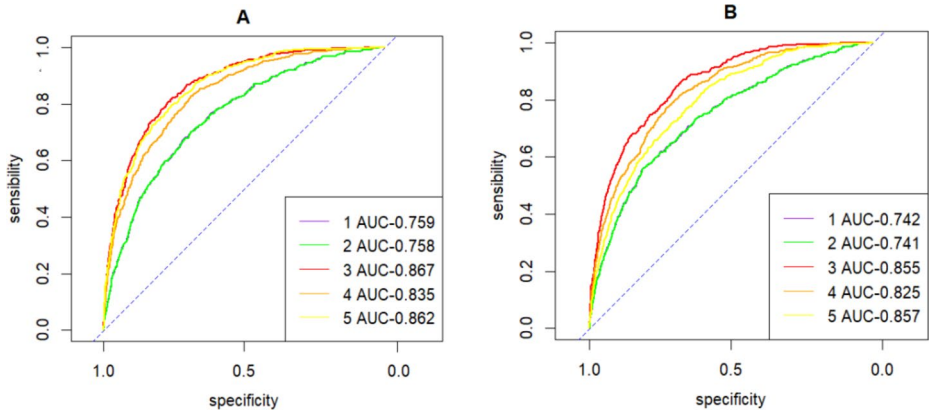


Fig. 2. ROC curve of anthropometric indicators predicting SA and SO occurrence. (**A**) Presents the ROC curve for the prediction of SA occurrence in the training group based on human anthropometric measurements; (**B**) exhibits the ROC curve for the prediction of SO occurrence in the training group.

Receiver-operating characteristic (ROC) curve analysis and optimum thresholds for anthropometric indices

The prognostic models for diverse anthropometric indicators demonstrated commendable performance (AUC > 0.7), with each model showcasing discernible predictive capability. Furthermore, the AUC value in the validation cohort corresponded with the training cohort, affirming the sustained predictive efficacy of the models in the validation group. In the SA training samples, the BRI indicator yielded the most robust predictive effect for SA (AUC = 0.867), closely pursued by WHtR (AUC = 0.862). Conversely, in the SO training samples, WHtR emerged as the most potent predictor for SO (AUC = 0.857), followed by BRI (AUC = 0.855). Noteworthy is the comparatively suboptimal predictive efficacy of the ABSI and LBSIZ models for SA and SO (Fig. 2).

The evaluation scores for each indicator were notably elevated, with the BRI metric exhibiting the highest Youden index, implicating its relatively heightened discriminatory ability for SA and SO. The sensitivity and specificity of the varied anthropometric indicators in predicting the onset of SA and SO are featured in Table 4.

Curves 1 to 5 correspond to the following anthropometric indices: ABSI, LBSIZ, BRI, BMI, WHtR.

Anthropometric indicators were used to predict SA and SO data derived from BIA measurements, and the results showed that the effect of each indicator model was good (AUC > 0.8) (Fig. 3).

Dose-response relationship of anthropometric indicators

Restrictive cubic spline visualization reveals a notable phenomenon wherein each anthropometric index exhibits a J-shaped correlation with the propensity for obstructive sleep apnea and snoring incidences, with a statistical significance of $P < 0.0001$ (Fig. 4).

For instance, taking the body roundness index (BRI) as a case in point, the risk of SA is minimized at a BRI value of 3.6, while the risk of SO reaches its nadir at a BRI of 3.7. Notably, any deviation from these optimal BRI values will result in an escalated risk of SA and SO occurrences.

Variable	Index	AUC	95%CI	Jorden index	Sensitivity	Specificity
Sarcopenia	ABSI	0.759	(0.74, 0.78)**	0.40	0.68	0.72
	LBSIZ	0.758	(0.74, 0.78)**	0.39	0.67	0.72
	BRI	0.867	(0.85, 0.88)**	0.58	0.81	0.77
	BMI	0.835	(0.82, 0.85)**	0.52	0.83	0.69
	WHtR	0.862	(0.85, 0.88)**	0.55	0.79	0.76
Sarcopenic Obesity	ABSI	0.742	(0.72, 0.76)**	0.37	0.56	0.81
	LBSIZ	0.741	(0.72, 0.76)**	0.37	0.56	0.81
	BRI	0.855	(0.84, 0.87)**	0.55	0.85	0.70
	BMI	0.825	(0.81, 0.84)**	0.50	0.77	0.73
	WHtR	0.857	(0.84, 0.87)**	0.54	0.83	0.71

Table 4. Assessed value of various anthropometric indicators on SA and SO occurrence. *ABSI* a body shape index, *BRI* body roundness index, *BMI* body mass index, *WHtR* waist-height ratio, *LBSIZ* the z-score of the log-transformed ABSI. $^{**}P < 0.001$.

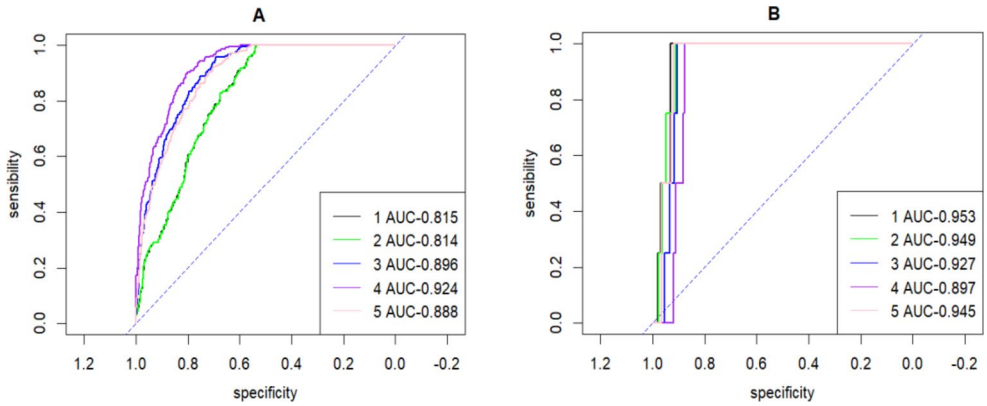


Fig. 3. ROC curve of anthropometric indicators predicting BIA data. (**A**) Presents the ROC curve for the prediction of SA occurrence in the BIA data based on human anthropometric measurements; (**B**) exhibits the ROC curve for the prediction of SO occurrence in the BIA data. Curves 1 to 5 correspond to the following anthropometric indices: ABSI, LBSIZ, BRI, BMI, WHtR.

Within the cohort of individuals aged 40, a stratified analysis by sex among the middle-income group unveils a sexual disparity in the incidence rates of SA and SO, with a higher prevalence observed among males compared to females. However, with the advancement of age, the prevalence rates of SA and SO among females increased at a faster pace than those of males. Notably, at a BRI value of 3.6, the risk of SA is mitigated for both sexes, whereas for females, a BRI range between 2.3 and 4.4 signifies a risk of SA lower than unity. Similarly, at a BRI value of 3.7, the risk of SO is minimized for both sexes, while for females, a BRI range between 2.3 and 5.0 corresponds to an SO risk below one (Fig. 5).

Discussion

This investigation reveals that hypertension and high cholesterol levels are not significantly associated with SA and SO, statistically speaking. Contrary to research suggesting a linkage between hypertension, high cholesterol, and the incidence of SA and SO[54–57], the findings of the present study lends support to an opposing hypothesis[58]. The discrepancy in outcomes may be attributed to variations in the study cohort and the metrics employed for assessment, which could account for the observed inconsistency in results[58].There is a positive associations between all anthropometric measurements and the incidence of SA and SO. After controlling for all covariates, the correlation between WHtR and SA, SO was found to be stronger, indicating a potential association between elevated WHtR and increased risk of metabolic diseases[59]. The ROC curve was employed in the construction of the prediction model, demonstrating excellent performance across all models. Notably, the Body Roundness Index (BRI) and the WHtR, metrics utilized to quantify central adiposity[60,61], have demonstrated a high degree of validity in these analyses.

Patients afflicted with SA and SO frequently present with abdominal obesity and visceral fat accumulation[34,62], Differential distribution of body fat exerts a significant impact on metabolic function[63]. An excessive accumulation of adipose tissue disrupts homeostasis and triggers a cascade of inflammatory responses. Moreover, the interplay between adipokines and myokines, along with a dysregulated immune-inflammatory response, interferes with the regenerative processes of skeletal muscle, potentially leading to comprehensive sarcopenia and sarcopenic obesity[64,65]. Diverging from the outcomes of prior research[40], the predictive efficacy of the ABSI and LBSIZ

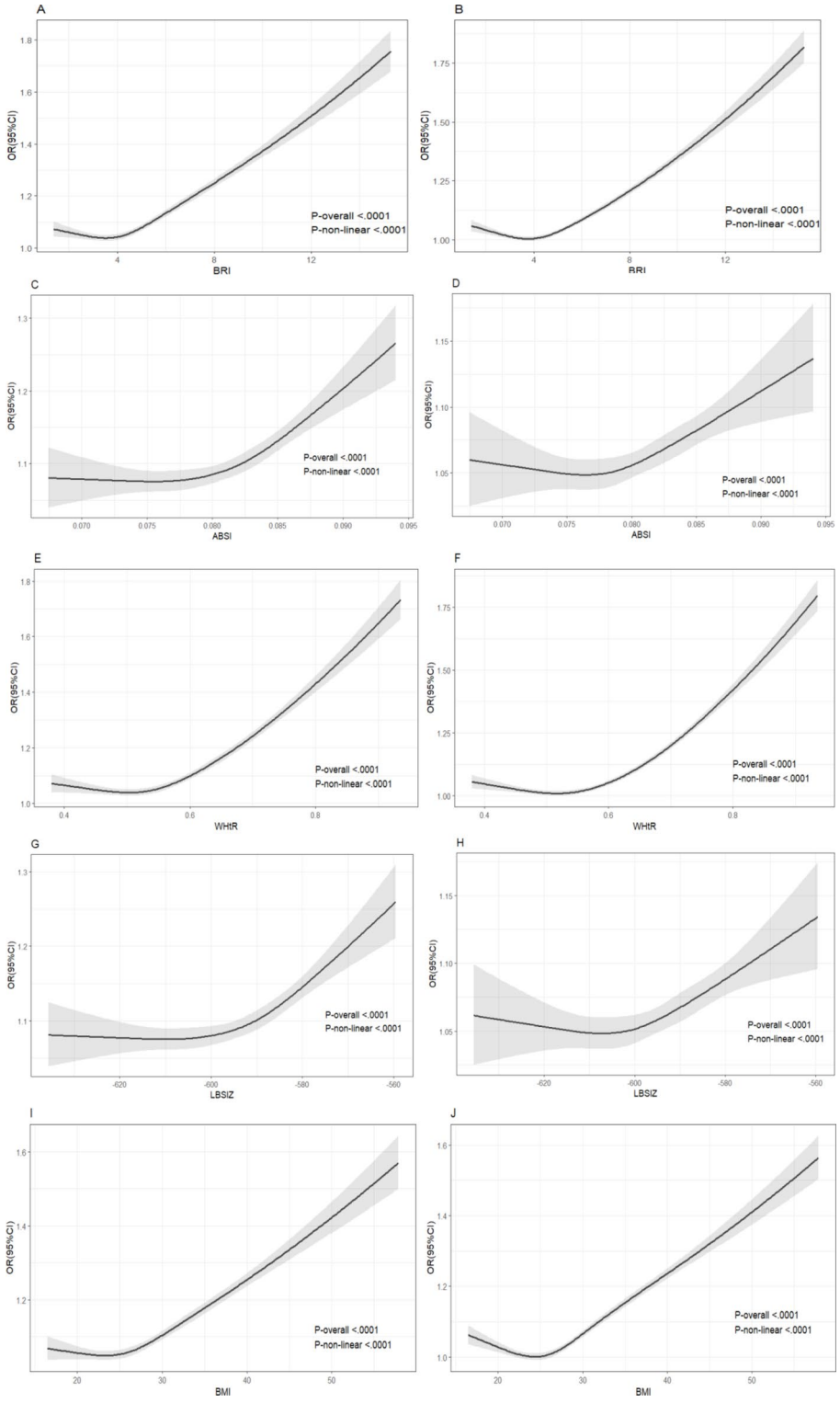


Fig. 4. Dose-effect relationship between anthropometric indexes and SA and SO occurrence. (**A**,**C**,**E**,**G**,**I**) are the dose-effect relationship between BRI, ABSI, WHtR, LBSIZ, BMI and SA occurrence, respectively. (**B**,**D**,**F**,**H**,**J**) are the dose-effect relationship between BRI, ABSI, WHtR, LBSIZ, BMI and SO occurrence, respectively.

models appears to be relatively modest. The potential explanation for this discrepancy could be that SA and SO are chronic conditions associated with aging, and the ABSI and LBSIZ may not be robust predictors for chronic illnesses[66], these indices may serve a supplementary role in the initial evaluation of human tissue distribution[24].

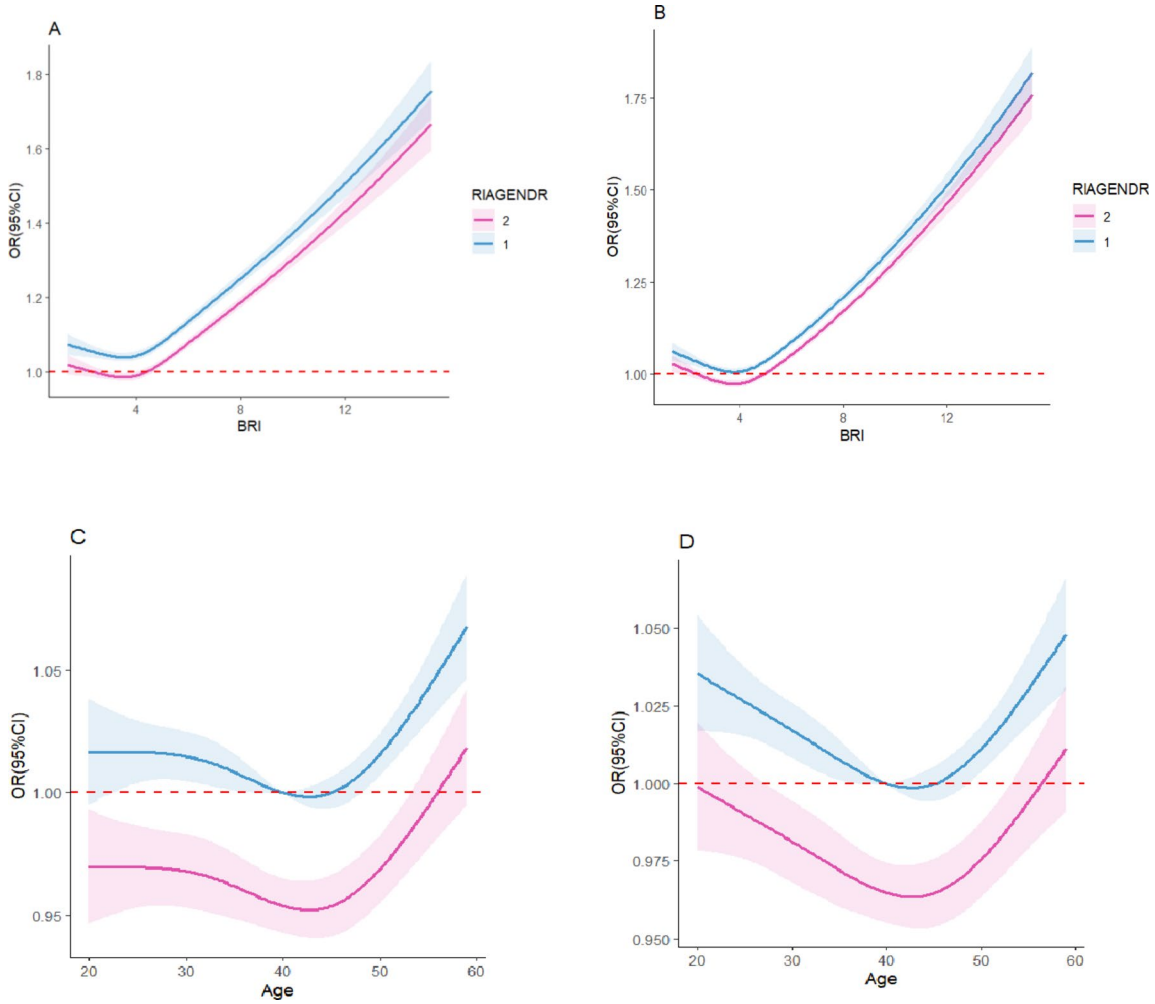


Fig. 5. Dose-effect relationship between SA and SO risk by sex 1 for male and 2 for female. (**A**) Is the dose-effect relationship of SA risk of different BRI index, (**B**) is the dose-effect relationship of SO risk of different BRI index, (**C**) is the dose-effect relationship of SA risk of different age, and (**D**) is the dose-effect relationship of SO risk of different age.

The RCS curve illustrates that each anthropometric index displays a J-shaped curve with respect to sarcopenia (SA) and osteopenia (SO), suggesting the presence of an optimal threshold for the anthropometric index where the prevalence of SA and SO is minimized. Any deviation from this threshold will augmented the susceptibility to disease development. The present investigation indicate that among the 40-year-old demographic, males exhibit a higher incidence of disease. Beginning at the age of 30, there is an annual decrease of approximately 1% in testosterone levels among males[67]. The elevation of visceral adiposity resultant from this decline in testosterone can elicit a catabolic influence on muscle tissue through the action of pro-inflammatory cytokines[68]. In females, oestrogen is instrumental in promoting growth and maintaining muscle mass in females, with estradiol specifically mitigating inflammatory stress-related damage to skeletal muscle[69,70]. Nonetheless, it is pertinent to highlight that the decline in estrogen levels post-menopause is often associated with a propensity for increased subcutaneous fat deposition. This shift may conducive to enhanced adiposity in postmenopausal women, thereby potentially predisposing them to the onset of SA and SO[71,72]. Individuals in middle age and older adulthood, irrespective of biological sex, should prioritize the regular assessment of their skeletal muscle health, assessing health risks, ensuring adequate intake of protein and energy in their daily lives, and enhancing physical activity as well as adhering to a regimen of regular exercise.

This investigation proposes that when assessing primary medical care and physical fitness, the consideration of BMI alone is insufficient. It is imperative to also take into account the results of body measurement indexes such as BRI and WHtR, as their relevance to the risk of SA and SO is more pronounced. The practical utility of these indices is underscored by their facilitation in measurement, capacity for direct computation, and high level of precision.

Despite the study's strength in its substantial and representative sample size, several limitations must be acknowledged. Primarily, the cross-sectional nature of this research precludes a thorough investigation into causal relationships. Secondly, although the analysis has been adjusted for a selection of potential confounding variables, the possibility of residual confounding remains. Furthermore, the study fails to address the mechanistic

association between muscle loss syndrome and osteoporosis[73,74], a link compounded by hormone levels impacting not only muscle tissue but also closely intertwining with mineral metabolism in bones[75]. However, this study neglects to delve into the complications of SA and SO, such as osteoporosis, due to the presence of confounding factors. In addition, owing to the lack of data pertaining to calf circumference (CC) and mid-upper arm circumference (AC) within the database, this investigation was precluded from conducting a comparative analysis of the predictive efficacy of the anthropometric indices under consideration with those methods. Lastly, given that the study's population primarily comprises Americans, further research is requisite to explore the association between body measurement indexes and SA and SO in Asian populations.

Conclusions

A variety of anthropometric measurements demonstrate significant associations with adult sarcopenia and sarcopenic obesity, representing valuable predictive factors. Notably, the body roundness index (BRI) and the waist-to-height ratio (WHtR) emerge as particularly promising anthropometric indicators. Furthermore, a dose-response analysis reveals the optimal thresholds for these measurements, corresponding to the lowest risk of developing sarcopenia and sarcopenic obesity. Consequently, it is prudent for individuals reaching middle age and beyond to endeavor towards vigilant surveillance of their muscular health and concomitant health risks. Embracing a regimen that ensures adequate daily protein and energy intake, in conjunction with a commitment to regular physical exercise, stands pivotal in the endeavor to safeguard overall well-being.

Data availability

All data are available at the National Health and Nutrition Examination Survey (https://www.cdc.gov/nchs/nhanes/index.htm).In addition, the data files used have been provided in the "related files".

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W. X. is responsible for writing the topic of the paper, C. Z. and Z. Y. provide technical help on the paper, and Professor Ren Hong provides guidance and assistance in all aspects.

Declarations

Competing interests
The authors declare no competing interests.

Ethical approval
This study was approved by the Ethics Committee of Beijing Sport University, all methods are carried out in accordance with the relevant guidelines and regulations.

Informed consent
This paper is based on an open database analysis of existing data, and all participants had signed informed consent forms at the time of data collection.

Additional information
Correspondence and requests for materials should be addressed to H.R.

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